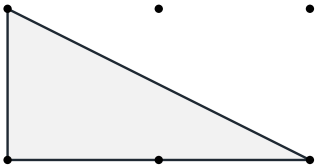
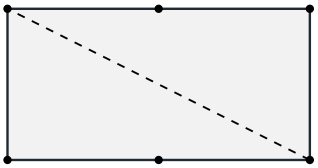
Join dots with straight sides to make one closed outline. Sides cannot cross or touch except at neighboring corners. Leave no holes until Problem 9. The area of one grid square is 1. Dots are 20 mm apart; print at 100%.

Input



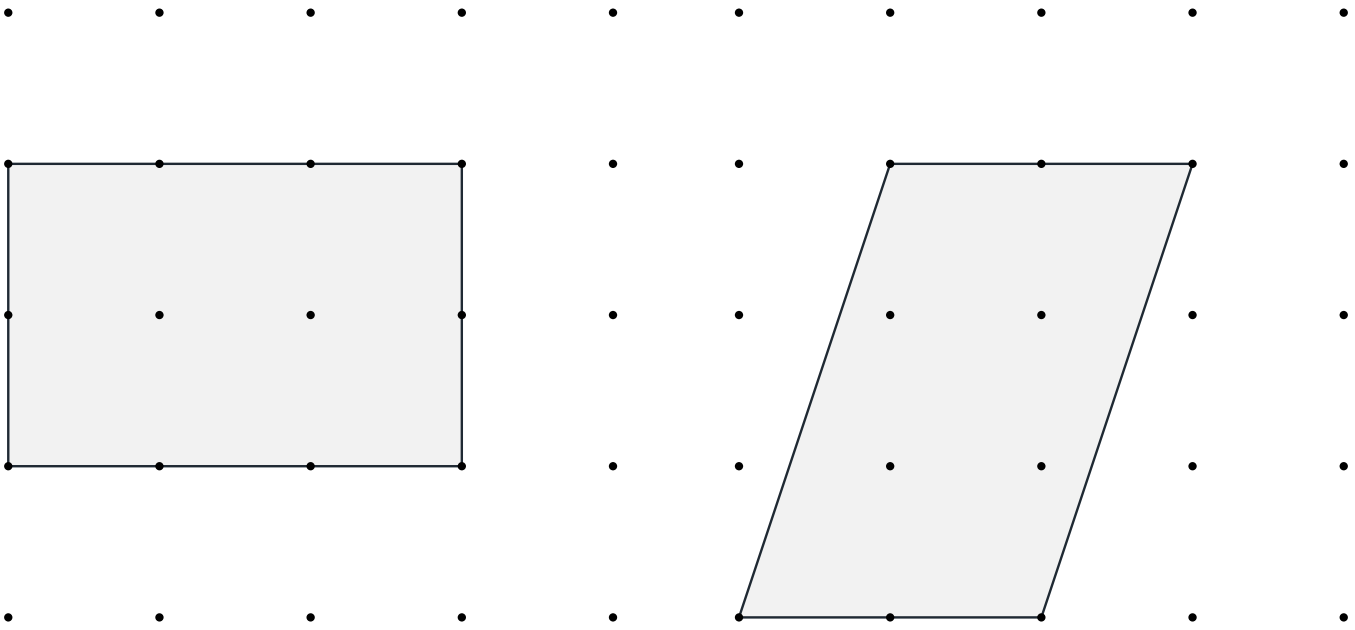
Two copies joined



Output

Rectangle area: 2
One copy's area: 1

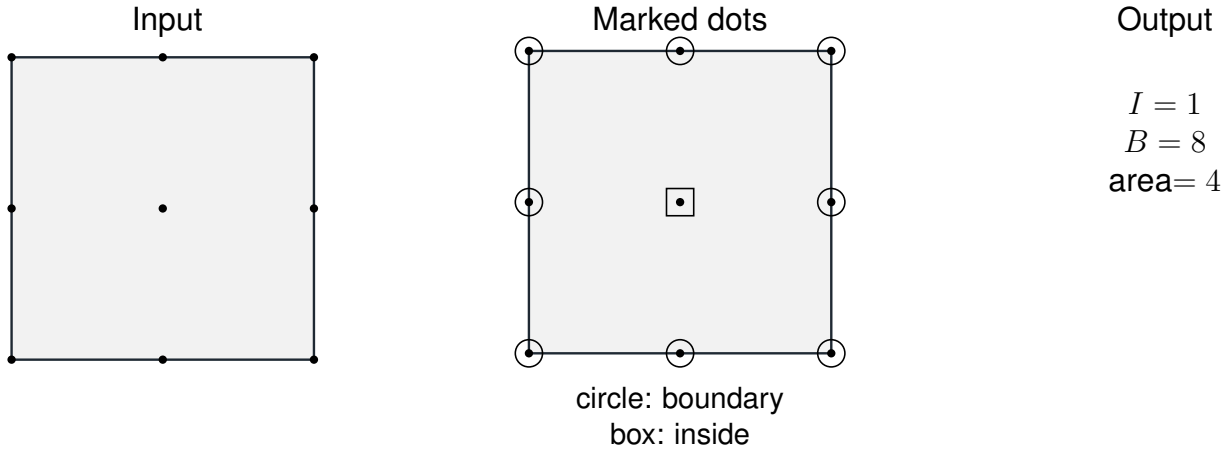
Problem 1: Find the exact area of each shape. Show a way to check each area with a cut or a second copy.



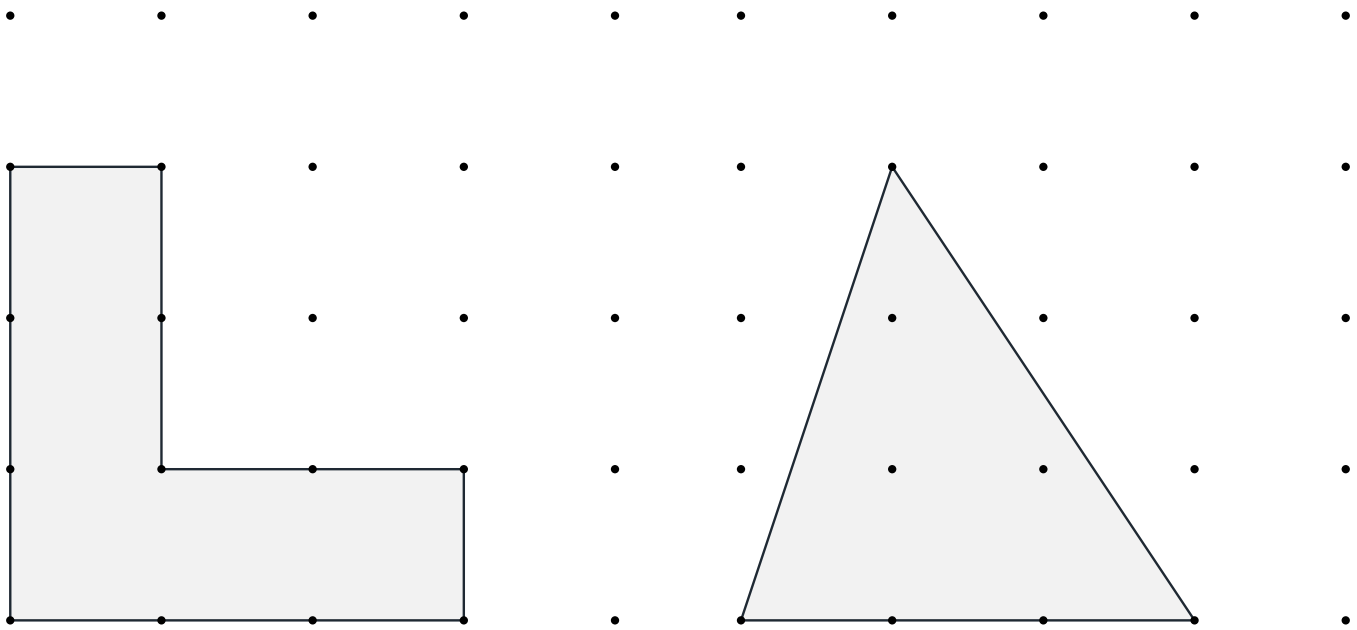
area=_____

area=_____

I counts dots strictly inside. B counts every dot on the outline, including dots between corners. Count each dot once.



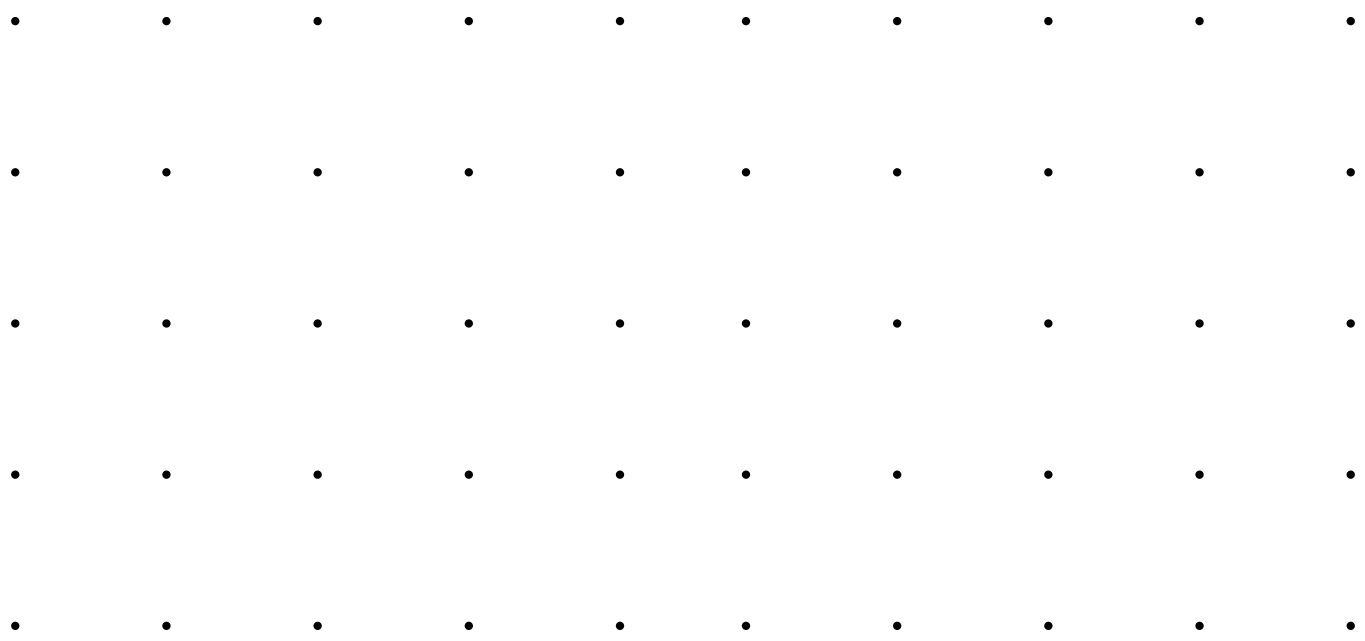
Problem 2: For each shape, mark the dots you count and make an $I, B,$ area record. Check area with pieces, cuts or copies.



$I =$_____ $B =$_____ area=_____

$I =$_____ $B =$_____ area=_____

Problem 3: Can two different shapes have the same I and B? Make a triangle and a four-sided shape, each with $I = 2$ and $B = 6$. Find their areas using cuts or copies.



$I =$_____ $B =$_____ area=_____ $I =$_____ $B =$_____ area=_____

Problem 4: Find a rule that predicts area from I and B. Use all your records. Draw a new shape that could defeat your rule.



$I =$_____ $B =$_____ area=_____

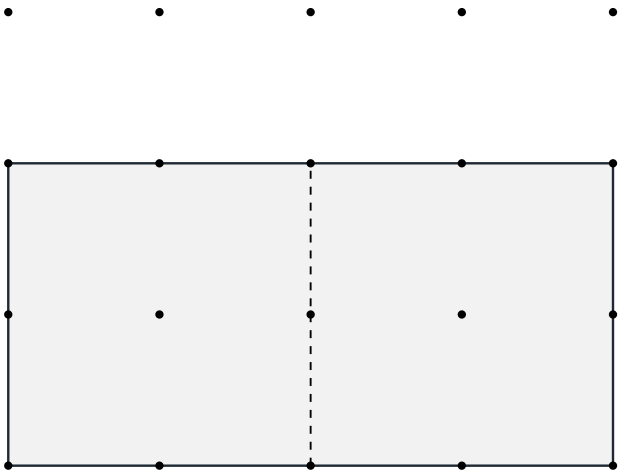
Problem 5: Pick's rule for the allowed polygons is

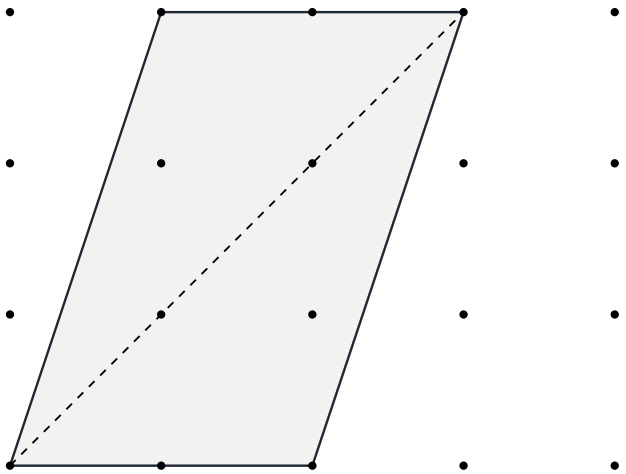
$$\text{area} = I + (\text{half of } B) - 1.$$

Make shapes of area 6 with as many different (I, B) records as you can. Check at least one area with pieces, cuts or copies.



Problem 6: Each dashed seam cuts a shape into two pieces. Make an $I, B,$ area record for each piece and for the whole shape. What changes when the seam disappears?





Left piece:
$I =$_____ $B =$_____ area=_____
Right piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____

Lower piece:
$I =$_____ $B =$_____ area=_____
Upper piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____